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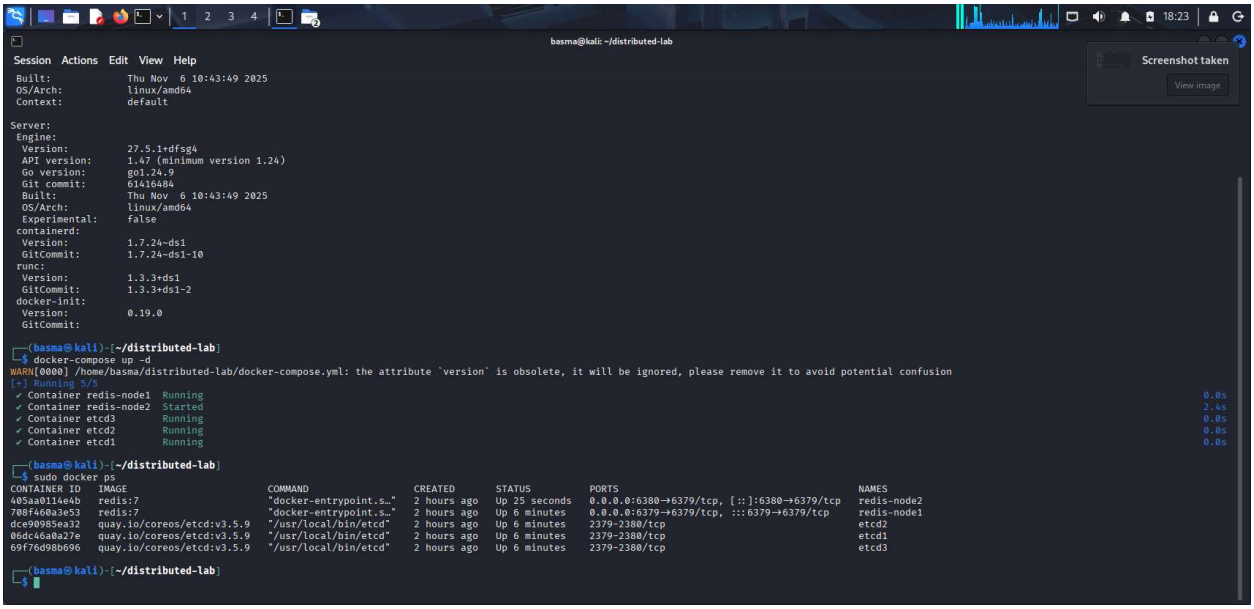
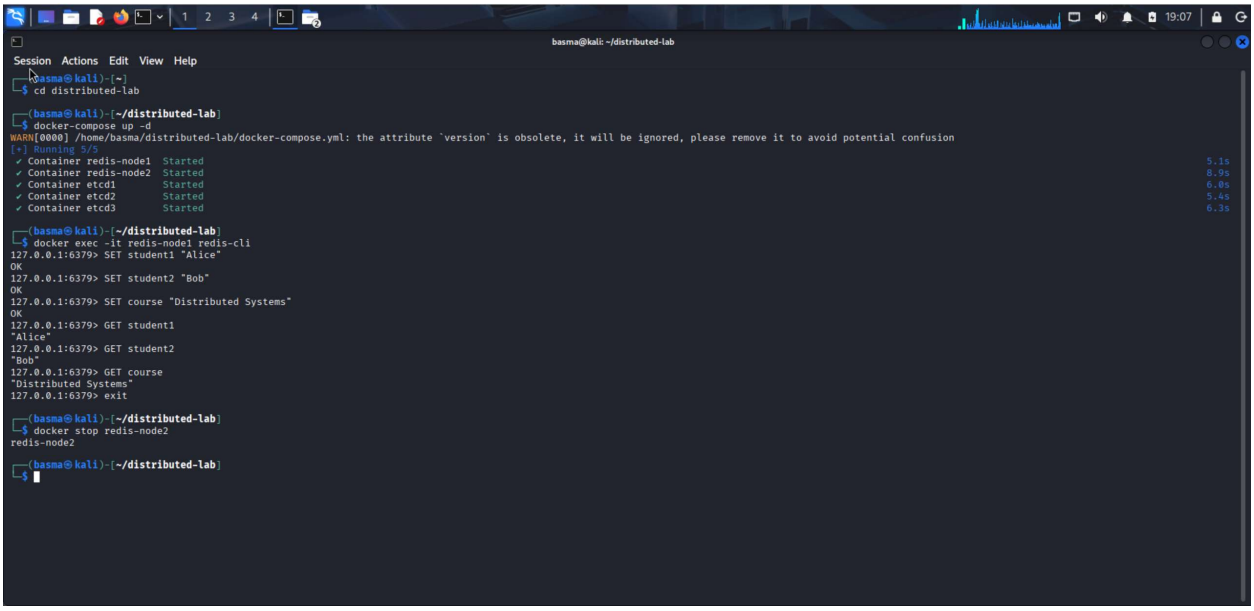
Group: SE3

Lab 2

1. Show Container Resources



2. Stop Container



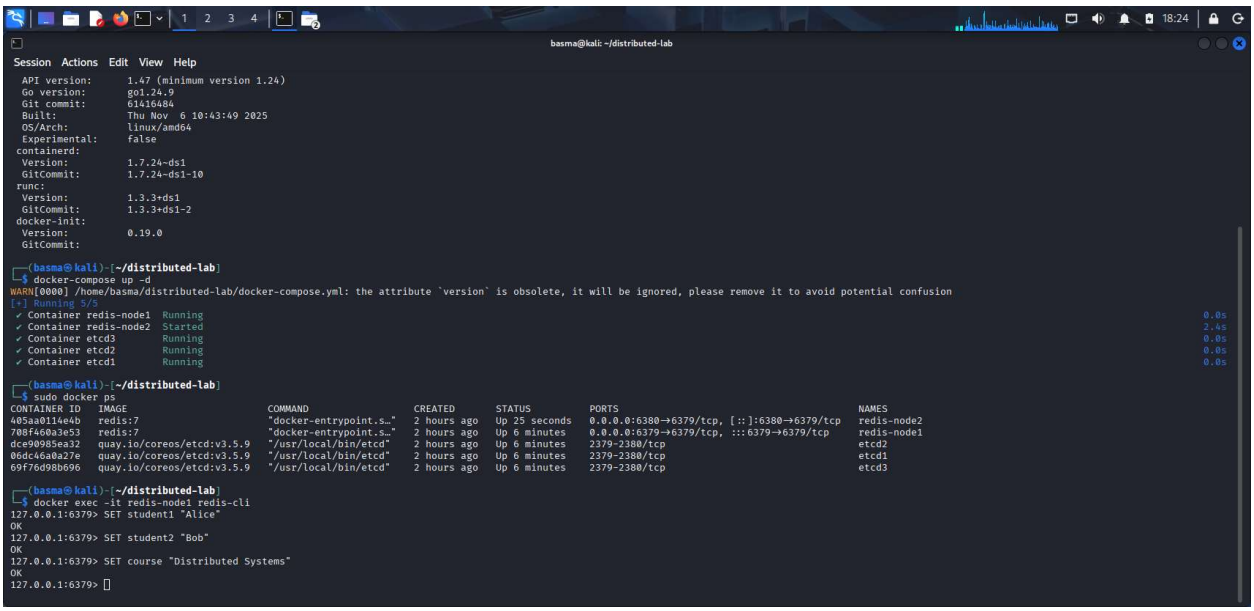
3. Remove Container

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker-compose up -d
WARN[0000] /home/basma/distributed-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 5/5
 ✔ Container redis-node1  Started                                                                                    5.1s
 ✔ Container redis-node2  Started                                                                                    8.9s
 ✔ Container etcd1        Started                                                                                    6.0s
 ✔ Container etcd2        Started                                                                                    5.4s
 ✔ Container etcd3        Started                                                                                    6.3s

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student1 "Alice"
OK
127.0.0.1:6379> SET student2 "Bob"
OK
127.0.0.1:6379> SET course "Distributed Systems"
OK
127.0.0.1:6379> GET student1
"Alice"
127.0.0.1:6379> GET student2
"Bob"
127.0.0.1:6379> GET course
"Distributed Systems"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker rm redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker ps
CONTAINER ID   IMAGE                       COMMAND                  CREATED       STATUS          PORTS                                          NAMES
708f460a3e53   redis:7                     "docker-entrypoint.s…"   3 hours ago   Up 10 minutes   0.0.0.0:6379->6379/tcp, :::6379->6379/tcp      redis-node1
dce90985ea32   quay.io/coreos/etcd:v3.5.9  "/usr/local/bin/etcd"    3 hours ago   Up 10 minutes   2379-2380/tcp                                  etcd2
06dc46a0a27e   quay.io/coreos/etcd:v3.5.9  "/usr/local/bin/etcd"    3 hours ago   Up 10 minutes   2379-2380/tcp                                  etcd1
69f76d98b696   quay.io/coreos/etcd:v3.5.9  "/usr/local/bin/etcd"    3 hours ago   Up 10 minutes   2379-2380/tcp                                  etcd3

┌──(basma㉿kali)-[~/distributed-lab]
└─$
```

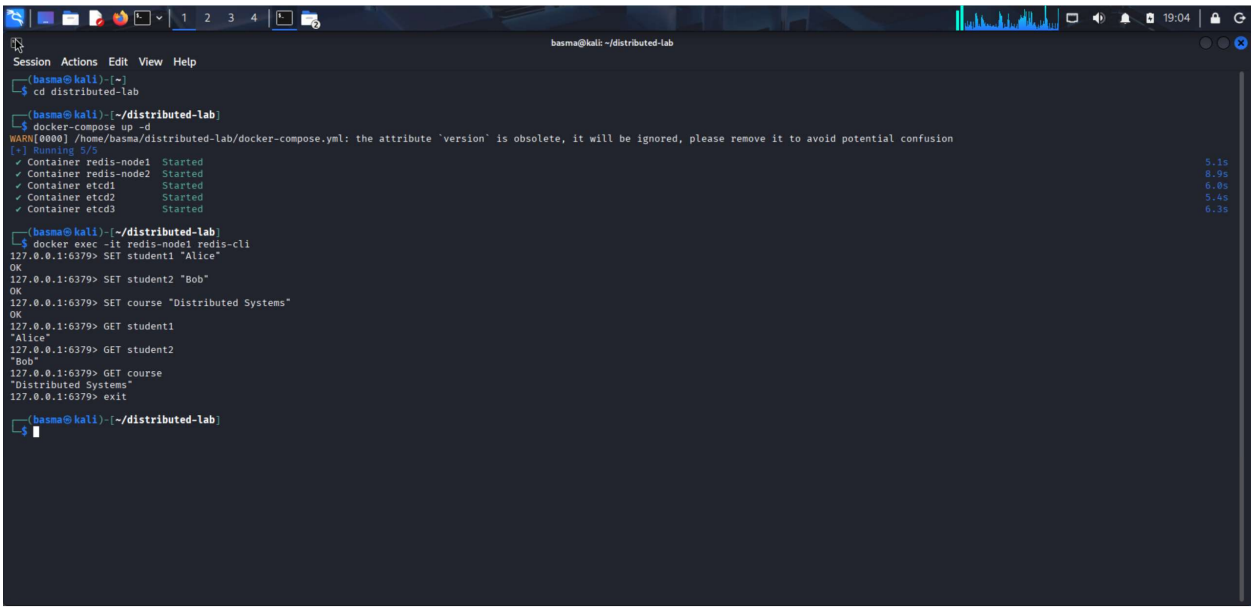
4. Start the services in the .yml or .yaml file.

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sudo docker version
[sudo] password for basma:
Client:
 Version:           27.5.1+dfsg4
 API version:       1.47
 Go version:        go1.24.9
 Git commit:        cab968b3
 Built:             Thu Nov  6 10:43:49 2025
 OS/Arch:           linux/amd64
 Context:           default

Server:
 Engine:
  Version:          27.5.1+dfsg4
  API version:      1.47 (minimum version 1.24)
  Go version:       go1.24.9
  Git commit:       61416484
  Built:            Thu Nov  6 10:43:49 2025
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          1.7.24-ds1
  GitCommit:        1.7.24-ds1-10
 runc:
  Version:          1.3.3+ds1
  GitCommit:        1.3.3+ds1-2
 docker-init:
  Version:          0.19.0
  GitCommit:

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker-compose up -d
WARN[0000] /home/basma/distributed-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 5/5
 ✔ Container redis-node1  Running                                                                                    0.0s
 ✔ Container redis-node2  Started                                                                                    2.4s
 ✔ Container etcd3        Running                                                                                    0.0s
 ✔ Container etcd2        Running                                                                                    0.0s
 ✔ Container etcd1        Running                                                                                    0.0s

┌──(basma㉿kali)-[~/distributed-lab]
└─$
```

5. Connect to Primary node and Write Data



6. Read it back from the Primary



7. Connect to Replica and read the same keys written to the Primary

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
  GitCommit:        1.3.3+ds1-2
 docker-init:
  Version:          0.19.0
  GitCommit:

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker-compose up -d
WARN[0000] /home/basma/distributed-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 5/5
 ✔ Container redis-node1  Running                                                                                    0.0s
 ✔ Container redis-node2  Started                                                                                    2.4s
 ✔ Container etcd3        Running                                                                                    0.0s
 ✔ Container etcd2        Running                                                                                    0.0s
 ✔ Container etcd1        Running                                                                                    0.0s

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sudo docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED       STATUS          PORTS                                          NAMES
405aa0114e4b   redis:7                      "docker-entrypoint.s…"   2 hours ago   Up 25 seconds   0.0.0.0:6380→6379/tcp, [::]:6380→6379/tcp      redis-node2
708f460a3e53   redis:7                      "docker-entrypoint.s…"   2 hours ago   Up 6 minutes    0.0.0.0:6379→6379/tcp, :::6379→6379/tcp        redis-node1
dce90985ea32   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd2
06dc46a0a27e   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd1
69f76d98b696   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd3

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student1 "Alice"
OK
127.0.0.1:6379> SET student2 "Bob"
OK
127.0.0.1:6379> SET course "Distributed Systems"
OK
127.0.0.1:6379>
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student1
"Alice"
127.0.0.1:6379> GET student2
"Bob"
127.0.0.1:6379> GET course
"Distributed Systems"
127.0.0.1:6379>
```

8. Stop the Replica (to simulate network failure)

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
WARN[0000] /home/basma/distributed-lab/docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 5/5
 ✔ Container redis-node1  Running                                                                                    0.0s
 ✔ Container redis-node2  Started                                                                                    2.4s
 ✔ Container etcd3        Running                                                                                    0.0s
 ✔ Container etcd2        Running                                                                                    0.0s
 ✔ Container etcd1        Running                                                                                    0.0s

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sudo docker ps
CONTAINER ID   IMAGE                        COMMAND                  CREATED       STATUS          PORTS                                          NAMES
405aa0114e4b   redis:7                      "docker-entrypoint.s…"   2 hours ago   Up 25 seconds   0.0.0.0:6380→6379/tcp, [::]:6380→6379/tcp      redis-node2
708f460a3e53   redis:7                      "docker-entrypoint.s…"   2 hours ago   Up 6 minutes    0.0.0.0:6379→6379/tcp, :::6379→6379/tcp        redis-node1
dce90985ea32   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd2
06dc46a0a27e   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd1
69f76d98b696   quay.io/coreos/etcd:v3.5.9   "/usr/local/bin/etcd"    2 hours ago   Up 6 minutes    2379-2380/tcp                                  etcd3

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student1 "Alice"
OK
127.0.0.1:6379> SET student2 "Bob"
OK
127.0.0.1:6379> SET course "Distributed Systems"
OK
127.0.0.1:6379>
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student1
"Alice"
127.0.0.1:6379> GET student2
"Bob"
127.0.0.1:6379> GET course
"Distributed Systems"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$
```

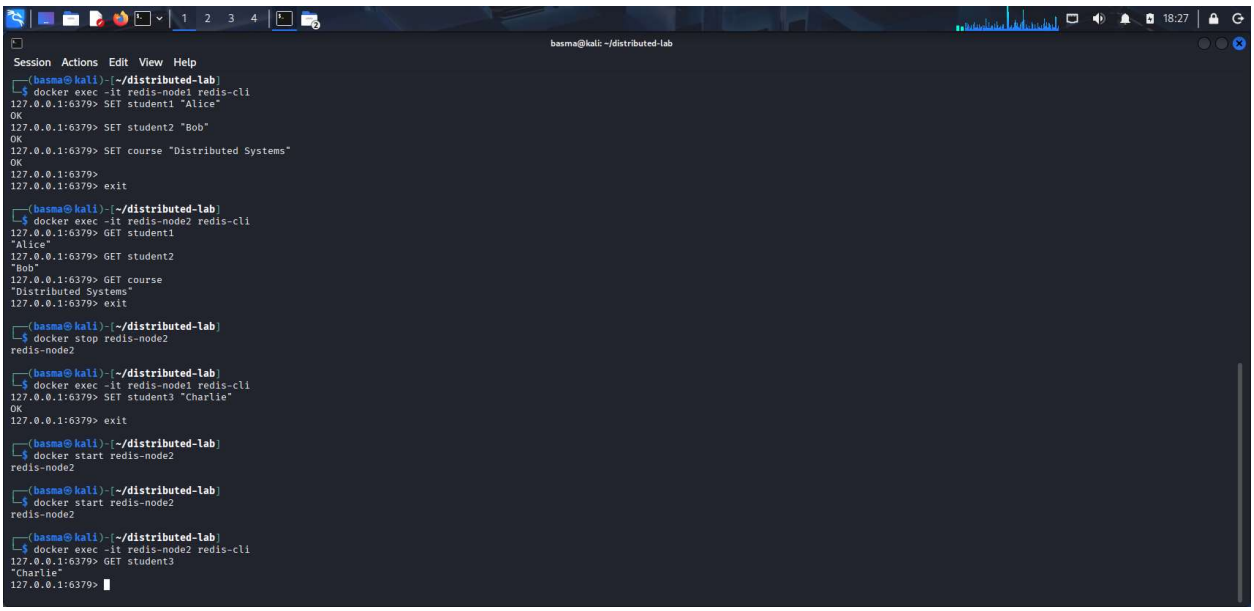
9. Write new data

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
 ✔ Container redis-node2  Started                                                                    2.4s
 ✔ Container etcd3        Running                                                                    0.0s
 ✔ Container etcd2        Running                                                                    0.0s
 ✔ Container etcd1        Running                                                                    0.0s

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sudo docker ps
CONTAINER ID   IMAGE                      COMMAND                  CREATED       STATUS          PORTS                                                   NAMES
405aa0114e4b   redis:7                    "docker-entrypoint.s…"   2 hours ago   Up 25 seconds   0.0.0.0:6380->6379/tcp, [::]:6380->6379/tcp             redis-node2
708f460a3e53   redis:7                    "docker-entrypoint.s…"   2 hours ago   Up 6 minutes    0.0.0.0:6379->6379/tcp, :::6379->6379/tcp               redis-node1
dce90985ea32   quay.io/coreos/etcd:v3.5.9 "/usr/local/bin/etcd"   2 hours ago   Up 6 minutes    2379-2380/tcp                                           etcd2
06dc46a0a27e   quay.io/coreos/etcd:v3.5.9 "/usr/local/bin/etcd"   2 hours ago   Up 6 minutes    2379-2380/tcp                                           etcd1
69f76d98b696   quay.io/coreos/etcd:v3.5.9 "/usr/local/bin/etcd"   2 hours ago   Up 6 minutes    2379-2380/tcp                                           etcd3

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student1 "Alice"
OK
127.0.0.1:6379> SET student2 "Bob"
OK
127.0.0.1:6379> SET course "Distributed Systems"
OK
127.0.0.1:6379>
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student1
"Alice"
127.0.0.1:6379> GET student2
"Bob"
127.0.0.1:6379> GET course
"Distributed Systems"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student3 "Charlie"
OK
127.0.0.1:6379>
```

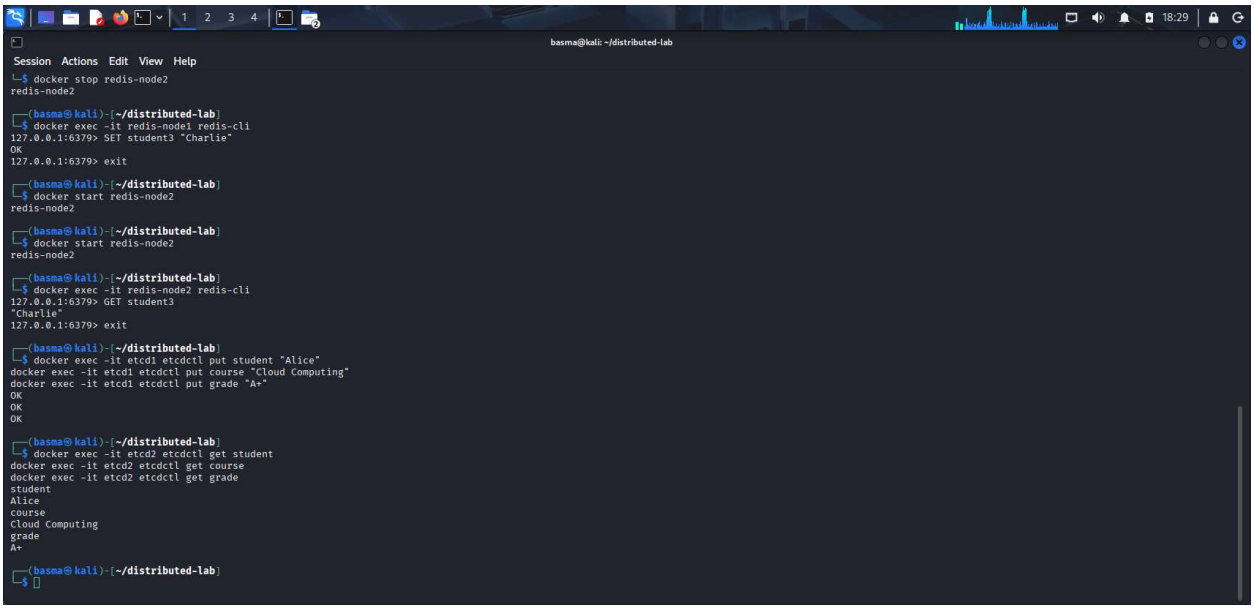
10. Restart the Stopped Replica and Reconnect to it

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student1 "Alice"
OK
127.0.0.1:6379> SET student2 "Bob"
OK
127.0.0.1:6379> SET course "Distributed Systems"
OK
127.0.0.1:6379>
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student1
"Alice"
127.0.0.1:6379> GET student2
"Bob"
127.0.0.1:6379> GET course
"Distributed Systems"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node1 redis-cli
127.0.0.1:6379> SET student3 "Charlie"
OK
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker start redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker start redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student3
"Charlie"
127.0.0.1:6379>
```

11. Read the newly entered data



12. Start the node cluster and Write Data



13. Read added data

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
└─$ docker start redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker start redis-node2
redis-node2

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student3
"Charlie"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd1 etcdctl put student "Alice"
docker exec -it etcd1 etcdctl put course "Cloud Computing"
docker exec -it etcd1 etcdctl put grade "A+"
OK
OK
OK

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl get student
docker exec -it etcd2 etcdctl get course
docker exec -it etcd2 etcdctl get grade
student
Alice
course
Cloud Computing
grade
A+
```

14. Check which node is the leader and Stop the leader

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it redis-node2 redis-cli
127.0.0.1:6379> GET student3
"Charlie"
127.0.0.1:6379> exit

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd1 etcdctl put student "Alice"
docker exec -it etcd1 etcdctl put course "Cloud Computing"
docker exec -it etcd1 etcdctl put grade "A+"
OK
OK
OK

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl get student
docker exec -it etcd2 etcdctl get course
docker exec -it etcd2 etcdctl get grade
student
Alice
course
Cloud Computing
grade
A+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd1 etcdctl endpoint status --write-out=table --cluster
+------------------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
|      ENDPOINT          |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+------------------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+
| http://etcd3:2379 | 876043ef79ada1ea |   3.5.9 |   20 kB |     false |      false |         3 |         15 |                 15 |        |
| http://etcd1:2379 | 99eab3685d8363a1 |   3.5.9 |   20 kB |      true |      false |         3 |         15 |                 15 |        |
| http://etcd2:2379 | dcb68c82481661be |   3.5.9 |   20 kB |     false |      false |         3 |         15 |                 15 |        |
+------------------------+------------------+---------+---------+-----------+------------+-----------+------------+--------------------+--------+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop etcd1
etcd1

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sleep 5

┌──(basma㉿kali)-[~/distributed-lab]
└─$
```

15. Check for New Leader and Write a new data

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
└─$ docker exec -it etcd1 etcdctl endpoint status --write-out=table --cluster
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
|      ENDPOINT          |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
| http://etcd3:2379 | 876043ef79ada1ea |   3.5.9 |   20 kB |     false |      false |         3 |         15 |                 15 |        |
| http://etcd1:2379 | 99eab3685d8363a1 |   3.5.9 |   20 kB |      true |      false |         3 |         15 |                 15 |        |
| http://etcd2:2379 | dcb68c82481661be |   3.5.9 |   20 kB |     false |      false |         3 |         15 |                 15 |        |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker stop etcd1
etcd1

┌──(basma㉿kali)-[~/distributed-lab]
└─$ sleep 5

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd1 etcdctl endpoint status --write-out=table --cluster
Error response from daemon: container 06dc46a0a27e78689ca9dd42dccc07a3a999e743a50b9bbebadc56bcedb51524 is not running

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl endpoint status --write-out=table --cluster
{"level":"warn","ts":"2026-03-04T23:32:17.9923222","logger":"etcd-client","caller":"v3@v3.5.9/retry_interceptor.go:62","msg":"retrying of unary invoker failed","target":"etcd-endpoints://0xc0002c4a80/127.0.0.1
:2379","attempt":0,"error":"rpc error: code = DeadlineExceeded desc = latest balancer error: last connection error: connection error: desc = \"transport: Error while dialing dial tcp: lookup etcd1 on 127.0.0.1
1:53: no such host\""}
Failed to get the status of endpoint http://etcd1:2379 (context deadline exceeded)
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
|      ENDPOINT          |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
| http://etcd3:2379 | 876043ef79ada1ea |   3.5.9 |   20 kB |      true |      false |         4 |         16 |                 16 |        |
| http://etcd2:2379 | dcb68c82481661be |   3.5.9 |   20 kB |     false |      false |         4 |         16 |                 16 |        |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl put state "Cairo"
OK
```

16. Start the stopped node and Read that new data from the started node

```
basma@kali: ~/distributed-lab
Session  Actions  Edit  View  Help
1:53: no such host\""}
Failed to get the status of endpoint http://etcd1:2379 (context deadline exceeded)
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
|      ENDPOINT          |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
| http://etcd3:2379 | 876043ef79ada1ea |   3.5.9 |   20 kB |      true |      false |         4 |         16 |                 16 |        |
| http://etcd2:2379 | dcb68c82481661be |   3.5.9 |   20 kB |     false |      false |         4 |         16 |                 16 |        |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl put state "Cairo"
OK

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker start etcd1
sleep 5
etcd1

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd1 etcdctl get state
docker exec -it etcd1 etcdctl get student
docker exec -it etcd1 etcdctl get course
docker exec -it etcd1 etcdctl get grade
state
Cairo
student
Alice
course
Cloud Computing
grade
A+

┌──(basma㉿kali)-[~/distributed-lab]
└─$ docker exec -it etcd2 etcdctl endpoint status --write-out=table --cluster
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
|      ENDPOINT          |        ID        | VERSION | DB SIZE | IS LEADER | IS LEARNER | RAFT TERM | RAFT INDEX | RAFT APPLIED INDEX | ERRORS |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+
| http://etcd3:2379 | 876043ef79ada1ea |   3.5.9 |   20 kB |      true |      false |         4 |         18 |                 18 |        |
| http://etcd1:2379 | 99eab3685d8363a1 |   3.5.9 |   20 kB |     false |      false |         4 |         18 |                 18 |        |
| http://etcd2:2379 | dcb68c82481661be |   3.5.9 |   20 kB |     false |      false |         4 |         18 |                 18 |        |
+------------------------+------------------+---------+--------+-----------+------------+-----------+------------+--------------------+--------+

┌──(basma㉿kali)-[~/distributed-lab]
└─$
```